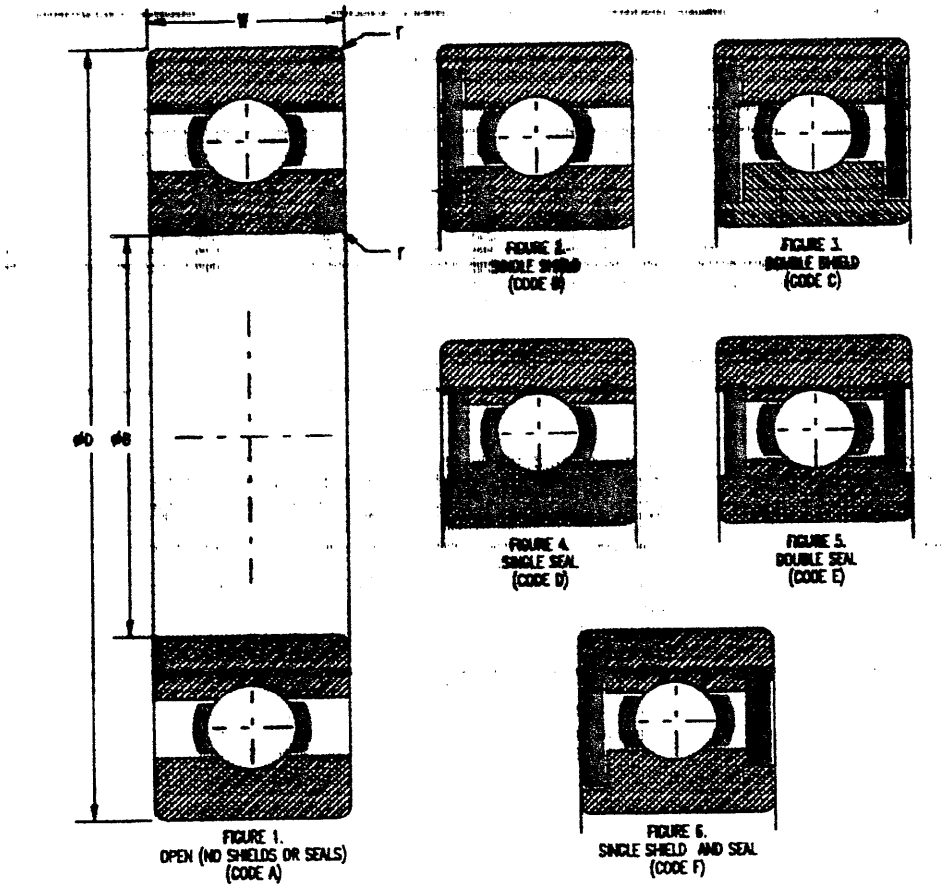
METRIC
FF-B-171/5
30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, SINGLE ROW, RADIAL, NON-FILLING SLOT, ANSI/AFBMA DIMENSION SERIES 03.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of **FF-B-171/GEN.**



BEARING CONFIGURATIONS

AMSC/NA

FSC 3110

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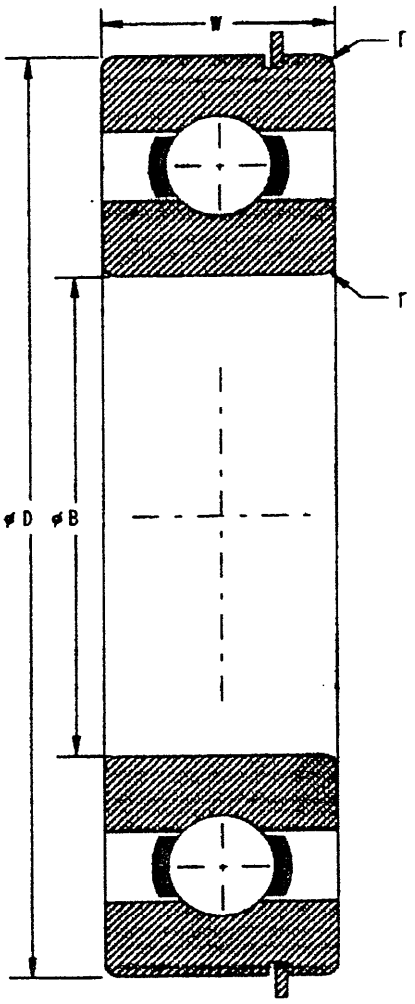


FIGURE 7.
OPEN SNAP RING
(CODE G)

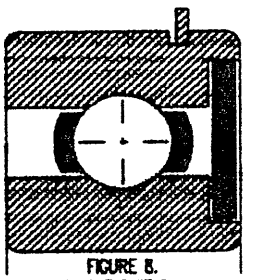


FIGURE 8.
SINGLE SHIELD-
ATTACHED ON SNAP
RING SIDE
(CODE H)

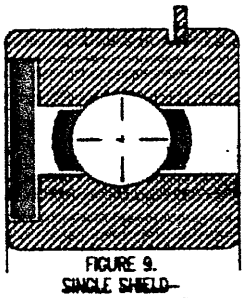


FIGURE 9.
SINGLE SHIELD-
ATTACHED ON SIDE
OPPOSITE SNAP RING
(CODE J)

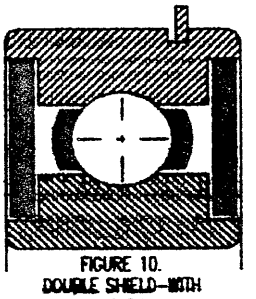


FIGURE 10.
DOUBLE SHIELD-WITH
SNAP RING
(CODE K)

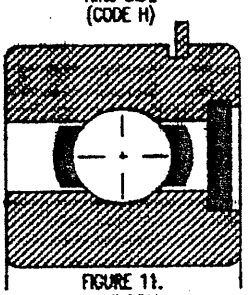


FIGURE 11.
SINGLE SEAL-
ATTACHED ON SNAP
RING SIDE
(CODE L)

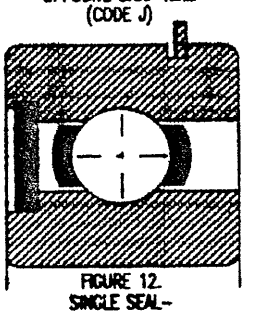


FIGURE 12.
SINGLE SEAL-
ATTACHED ON SIDE
OPPOSITE SNAP RING
(CODE M)

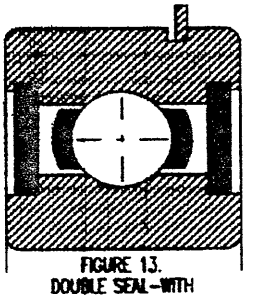


FIGURE 13.
DOUBLE SEAL-WITH
SNAP RING
(CODE N)

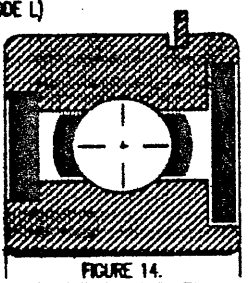


FIGURE 14.
SINGLE SHIELD AND SEAL-
SHIELD ON SNAP RING SIDE
(CODE P)

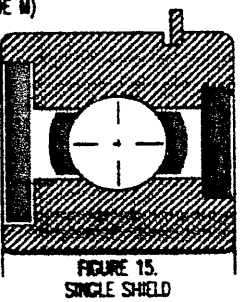


FIGURE 15.
SINGLE SHIELD
AND SEAL-SEAL
ON SNAP RING SIDE
(CODE R)

BEARING CONFIGURATIONS - (Continued)

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REQUIREMENTS:

1. **Boundary dimensions.** The boundary dimensions shall be specified in the part number by the size codes in Table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Shields, seals, and snap rings. Shields and seals shall be fabricated from any material unless otherwise specified and shall be removable or nonremovable type. Seals shall be impervious contact type. Snap rings shall be fabricated from ferrous material unless otherwise specified and shall have sufficient spring temper to set firmly in the ring grooves and shall be removable without the metal taking a permanent set. The snap ring and snap ring groove dimensions and their relative locations shall be in accordance with ANSI/AFBMA Standard 20. The number and positioning of shields, seals, and snap rings required shall be specified in the part number by the codes in table 111.

4. Radial **Internal Clearance.** The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

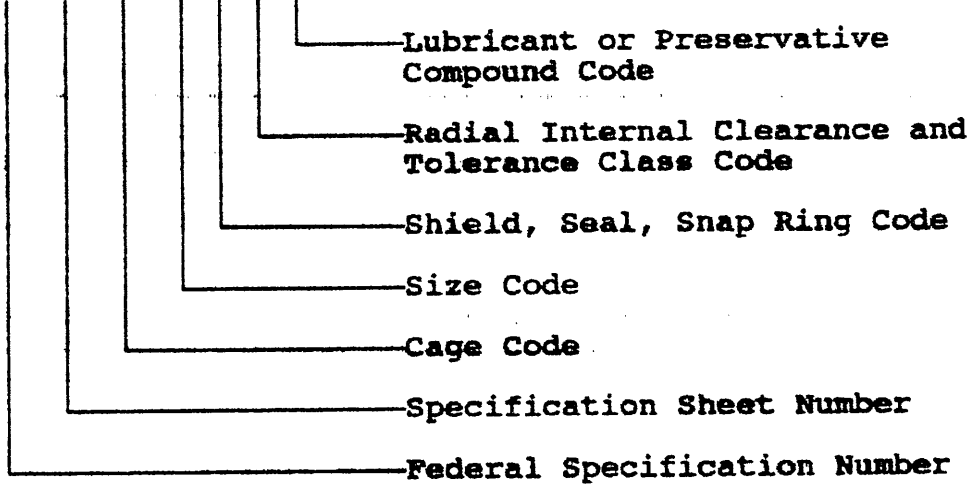
5. **Tolerance Class.** The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

6. **Lubricant** or Preservative **Compound.** Preservative compound shall be applied to all open bearings (no shields or seals) and to single closure bearings (mingle shield or single seal) that have no additional lubricant requirement. Thus, for these bearings the bearing part number shall include the "C" suffix. For double closure or single closure bearings that require lubricant the bearing part number shall include the "A" or "B" suffix. When grease is required, the bearing void (the airspace between the inner and outer ring of the assembled bearing) shall be 25 to 50 percent filled. The lubricant or preservative compound shall be specified in the part number by the codes in table V.

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7. **Part number.** The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, shields, seals, and snap rings, radial internal clearance and tolerance class, and lubricant or preservation compound.

EXAMPLE: FFB171/05 J 30 R 7 A



FFB171/05J30R7A indicates - Bearing, cage-pressed steel or molded plastic; bore l50mm; outside diameter 320mm; width 65mm; one shield and one seal (seal on snap ring side); radial internal clearance symbol 3 (greater than normal); tolerance class ABEC-5; grease in accordance with MIL-G-81322.

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TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic load rating (pounds)
00	10	35	11	0.6	15	1385
01	12	37	12	1.0	18	1670
02	15	42	13	1.0	21	1960
03	17	47	14	1.0	23	2340
04	20	52	15	1.1	27	2760
05	25	62	17	1.1	32	3670
06	30	72	19	1.1	37	4875
07	35	80	21	1.5	43.5	5780
08	40	90	23	1.5	48.5	7075
09	45	100	25	1.5	53.5	9160
10	50	110	27	2.0	60	10730
11	55	120	29	2.0	65	12410
12	60	130	31	2.1	72	14200
13	65	140	33	2.1	77	16095
14	70	150	35	2.1	82	18090
15	75	160	37	2.1	87	19705
16	80	170	39	2.1	92	21360
17	85	180	41	3.0	99	23055
18	90	190	43	3.0	104	24780
19	95	200	45	3.0	109	26540
20	100	215	47	3.0	114	30055
21	105	225	49	3.0	119	31915
22	110	240	50	3.0	124	35595
24	120	260	55	3.0	134	35930
26	130	280	58	4.0	148	39865
28	140	300	62	4.0	158	43915
30	150	320	65	4.0	168	47680
32	160	340	68	4.0	178	48240
34	170	360	72	4.0	188	54610
36	180	380	75	4.0	198	61530
38	190	400	78	5.0	212	63985

1/ Size codes in Table I are industry bore codes.

2/ The minimum bearing chamber ($r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.

3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

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TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)

TABLE III. Codes for Shields, Seals, and Snap Rings.

Code	Shield, seal, and snap ring configuration
A	open (no shields or seals) - fig.1
B	single shield - fig.2
C	double shield - fig.3
D	single seal - fig.4
E	double seal - fig.5
F	single shield and seal - fig.6
G	open with snap ring - fig.7
H	single shield (attached on snap ring side) - fig.8
J	single shield (attached on side opposite snap ring)-fig.9
K	double shield with snap ring - fig.10
L	single seal (attached on snap ring side) - fig.11
M	single seal (attached on side opposite snap ring)-fig.12
N	double seal with snap ring - fig.13
P	single shield and seal (shield on snap ring side)-fig.14
R	single shield and seal (seal on snap ring side) - fig.15

TABLE IV. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (less than normal)	ABEC-1
2	Symbol 0 (normal)	
3	Symbol 3 (greater than normal)	
4	Symbol 4 (greater than symbol 3)	
5	Symbol 2 (less than normal)	ABEC-5
6	Symbol 0 (normal)	
7	Symbol 3 (greater than normal)	
8	Symbol 4 (greater than symbol 3)	

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TABLE V. Codes for Lubricant and Preservative Compound.

Code	Lubricant or Preservative Compound
A	Grease in accordance with MIL-G-81322
B	Grease in accordance with DOD-G-24508
C	Preservative compound in accordance with MIL-P-197

MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army - AT
Air Force - 11

Review Activities:
Army - AR, EA, MI
Navy - OS

User Activity:
Army - GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

PREPARING ACTIVITY':
DLA-IS

(Project 3110-0848-05)

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